

ACR42827

## Phosphorus pentabromide

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 五溴化磷  
Phosphorus pentabromide

**Cat No. :** 428270000; 428270250; 428271000  
**CAS No** 7789-69-7  
**Molecular Formula** Br<sub>5</sub> P

**Supplier** **UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Solid

**Appearance**  
Yellow - Brown

**Odor**  
No information available

**Emergency Overview**  
Causes severe skin burns and eye damage. Reacts violently with water. Moisture sensitive.

#### Classification of the substance or mixture

|                                   |              |
|-----------------------------------|--------------|
| Skin Corrosion/Irritation         | Category 1 B |
| Serious Eye Damage/Eye Irritation | Category 1   |

#### Label Elements



**Signal Word**

**Danger**

**Hazard Statements**

## Phosphorus pentabromide

H314 - Causes severe skin burns and eye damage

**Precautionary Statements****Prevention**

P271 - Use only outdoors or in a well-ventilated area

P264 - Wash face, hands and any exposed skin thoroughly after handling

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P310 - Immediately call a POISON CENTER or doctor

P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Reacts violently with water.

**Health Hazards**

Corrosive. Causes skin and eye burns. Causes serious eye damage.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Reacts violently with water. Is not likely mobile in the environment. Reacts violently with water.

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component               | CAS No    | Weight % |
|-------------------------|-----------|----------|
| Phosphorus pentabromide | 7789-69-7 | >95      |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

## Phosphorus pentabromide

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Reacts violently with water.

**Protective Equipment and Precautions for Firefighters**

Thermal decomposition can lead to release of irritating gases and vapors. As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Avoid dust formation.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Do not expose spill to water.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not allow contact with water. Avoid dust formation. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

**Storage**

Keep away from water or moist air. Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Store under an inert atmosphere.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

## Phosphorus pentabromide

## Exposure Controls

## Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

## Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Neoprene       |                                   |                 |             |                       |
| Natural rubber |                                   |                 |             |                       |
| PVC            |                                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                 |                               |                                                   |
|---------------------------------|-------------------------------|---------------------------------------------------|
| <b>Appearance</b>               | Yellow - Brown                |                                                   |
| <b>Physical State</b>           | Solid                         |                                                   |
| <b>Odor</b>                     | No information available      |                                                   |
| <b>Odor Threshold</b>           | No data available             |                                                   |
| <b>pH</b>                       | No information available      |                                                   |
| <b>Melting Point/Range</b>      | 105 - 107 °C / 221 - 224.6 °F |                                                   |
| <b>Softening Point</b>          | No data available             |                                                   |
| <b>Boiling Point/Range</b>      | No information available      |                                                   |
| <b>Flash Point</b>              | No information available      | <b>Method -</b> No information available<br>Solid |
| <b>Evaporation Rate</b>         | Not applicable                |                                                   |
| <b>Flammability (solid,gas)</b> | No information available      |                                                   |

**SAFETY DATA SHEET****Phosphorus pentabromide**

|                                                |                             |       |
|------------------------------------------------|-----------------------------|-------|
| <b>Explosion Limits</b>                        | No data available           |       |
| <b>Vapor Pressure</b>                          | No data available           |       |
| <b>Vapor Density</b>                           | Not applicable              | Solid |
| <b>Specific Gravity / Density</b>              | No data available           |       |
| <b>Bulk Density</b>                            | No data available           |       |
| <b>Water Solubility</b>                        | Reacts violently with water |       |
| <b>Solubility in other solvents</b>            | No information available    |       |
| <b>Partition Coefficient (n-octanol/water)</b> |                             |       |
| <b>Autoignition Temperature</b>                | No data available           |       |
| <b>Decomposition Temperature</b>               | No data available           |       |
| <b>Viscosity</b>                               | Not applicable              | Solid |
| <b>Explosive Properties</b>                    | No information available    |       |
| <b>Oxidizing Properties</b>                    | No information available    |       |
| <b>Molecular Formula</b>                       | Br5 P                       |       |
| <b>Molecular Weight</b>                        | 430.49                      |       |

**SECTION 10. STABILITY AND REACTIVITY**

|                                         |                                                                                                                 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Reacts violently with water. Moisture sensitive.                                                                |
| <b>Hazardous Reactions</b>              | Reacts violently with water.                                                                                    |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                        |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Exposure to moist air or water. Exposure to moisture. Avoid dust formation. |
| <b>Materials to avoid</b>               | Strong oxidizing agents. Acids. Strong bases. Alcohols. Water.                                                  |
| <b>Hazardous Decomposition Products</b> | Oxides of phosphorus.                                                                                           |

**SECTION 11. TOXICOLOGICAL INFORMATION**

|                                               |                                                             |
|-----------------------------------------------|-------------------------------------------------------------|
| <b>Product Information</b>                    | No acute toxicity information is available for this product |
| <b>(a) acute toxicity;</b>                    |                                                             |
| <b>(b) skin corrosion/irritation;</b>         | Category 1 B                                                |
| <b>(c) serious eye damage/irritation;</b>     | Category 1                                                  |
| <b>(d) respiratory or skin sensitization;</b> |                                                             |
| <b>Respiratory</b>                            | No data available                                           |
| <b>Skin</b>                                   | No data available                                           |
| <b>(e) germ cell mutagenicity;</b>            | No data available                                           |
| <b>(f) carcinogenicity;</b>                   | No data available                                           |
|                                               | There are no known carcinogenic chemicals in this product   |
| <b>(g) reproductive toxicity;</b>             | No data available                                           |

**SAFETY DATA SHEET****Phosphorus pentabromide**

|                                            |                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (h) STOT-single exposure;                  | No data available                                                                                                                                                                                                                                          |
| (i) STOT-repeated exposure;                | No data available                                                                                                                                                                                                                                          |
| Target Organs                              | No information available.                                                                                                                                                                                                                                  |
| (j) aspiration hazard;                     | Not applicable<br>Solid                                                                                                                                                                                                                                    |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                             |
| Symptoms / effects, both acute and delayed | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |

**SECTION 12. ECOLOGICAL INFORMATION**

|                                       |                                                                                                    |
|---------------------------------------|----------------------------------------------------------------------------------------------------|
| Ecotoxicity effects                   | Do not empty into drains. Reacts with water so no ecotoxicity data for the substance is available. |
| Persistence and Degradability         | No information available                                                                           |
| Persistence                           | Persistence is unlikely, based on information available.                                           |
| Degradability                         | Reacts with water.                                                                                 |
| Degradation in sewage treatment plant | Reacts violently with water.                                                                       |
| Bioaccumulative Potential             | Product does not bioaccumulate due to reaction with water                                          |
| Mobility in soil                      | Reacts violently with water Is not likely mobile in the environment                                |
| Endocrine Disruptor Information       | This product does not contain any known or suspected endocrine disruptors                          |
| Persistent Organic Pollutant          | This product does not contain any known or suspected substance                                     |
| Ozone Depletion Potential             | This product does not contain any known or suspected substance                                     |

**SECTION 13. DISPOSAL CONSIDERATIONS**

|                                     |                                                                                                                                                                                                               |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from Residues/Unused Products | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                        |
| Contaminated Packaging              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                     |
| Other Information                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms. |

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

|                      |                         |
|----------------------|-------------------------|
| UN-No                | UN2691                  |
| Proper Shipping Name | PHOSPHORUS PENTABROMIDE |
| Hazard Class         | 8                       |
| Packing Group        | II                      |

## Phosphorus pentabromide

**IMDG/IMO**

UN-No UN2691  
Proper Shipping Name PHOSPHORUS PENTABROMIDE  
Hazard Class 8  
Packing Group II

**IATA**

UN-No UN2691  
Proper Shipping Name PHOSPHORUS PENTABROMIDE  
Hazard Class 8  
Packing Group II

Special Precautions for User No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component               | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Phosphorus pentabromide | X                                                   | X                                       | X    | X     | 232-186-6 | X    | -   | X     | X    | X    | X    | KE-28718 |

**National Regulations****SECTION 16. OTHER INFORMATION**

Creation Date 28-Sep-2009  
Revision Date 11-Apr-2024  
Revision Summary Not applicable.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend**

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

Phosphorus pentabromide

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**WEL** - Workplace Exposure Limit**ACGIH** - American Conference of Governmental Industrial Hygienists**DNEL** - Derived No Effect Level**RPE** - Respiratory Protective Equipment**LC50** - Lethal Concentration 50%**NOEC** - No Observed Effect Concentration**PBT** - Persistent, Bioaccumulative, Toxic**TWA** - Time Weighted Average**IARC** - International Agency for Research on Cancer**PNEC** - Predicted No Effect Concentration**LD50** - Lethal Dose 50%**EC50** - Effective Concentration 50%**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road**OECD** - Organisation for Economic Co-operation and Development**BCF** - Bioconcentration factor**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**MARPOL** - International Convention for the Prevention of Pollution from Ships**ATE** - Acute Toxicity Estimate**VOC** - (Volatile Organic Compound)**Key literature references and sources for data**<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**